IT5437 — Computer Vision | Assignment 2

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GitHub: https://github.com/Kalana-Demintha-Vidanapathirana/258839R_IT5437_CV_Assignment_2.git

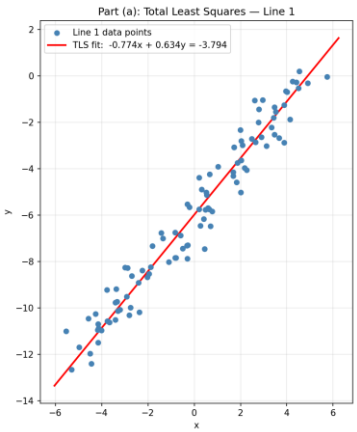
1. Line Fitting

(a) Total Least Squares — Line 1

TLS minimises orthogonal (perpendicular) distances to the line, unlike ordinary LS which minimises vertical residuals. The line is expressed in normal form ax+by=d. The unit normal (a,b) is the eigenvector of the scatter matrix U^TU corresponding to the smallest eigenvalue — equivalently the last row of V^T from SVD of the centred data matrix.

```
U = pts1 - np.mean(pts1, axis=0)  # centre data
_, _, Vt = np.linalg.svd(U.T @ U) # scatter matrix SVD
a, b = Vt[-1, 0], Vt[-1, 1]       # normal = last row of Vt
d    = a*mean[0] + b*mean[1]      # distance from origin
```

Result: a = -0.7736, b = 0.6337, d = -3.7942 | Mean perpendicular residual = 0.406

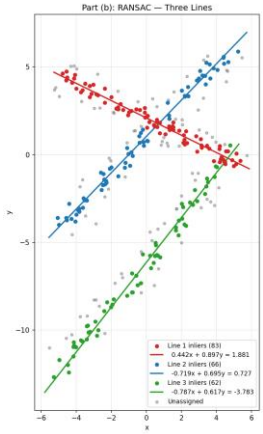


(b) RANSAC — Three Lines

All 300 mixed points (3 groups x 100) are passed to RANSAC. Each iteration: (1) sample 2 points, compute candidate line; (2) count inliers within threshold t=0.5; (3) refit with all inliers via TLS; (4) mask out inliers and repeat for the next line.

```
for i in range(3):
    model, mask = ransac_line(pts_all[remaining], t=0.5, max_iter=500)
    remaining[global_idx[mask]] = False  # remove claimed inliers
```

Line 1: 0.442x+0.897y=1.881 (83 inliers) Line 2: -0.719x+0.695y=0.727 (66 inliers) Line 3: -0.787x+0.617y=-3.783 (62 inliers). 89 points unassigned due to proximity to line intersections.



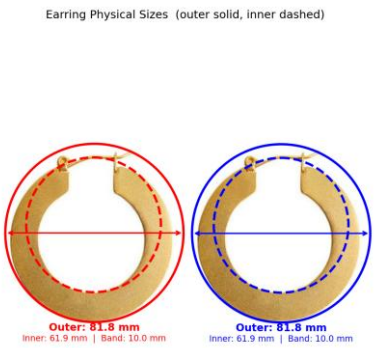
2. Physical Size of Earrings

The camera is perpendicular to the earring plane (object plane) at distance u=720 mm. Thin lens equation (1/f=1/u+1/v) with f=8 mm gives v=8.090 mm and magnification m=v/u=0.01124. Scale = pixel_size/m = 2.2e-3/0.01124 = 0.1958 mm/pixel.

```
f=8.0; p=2.2e-3; u=720.0
v = (f*u)/(u-f)   # 8.090 mm
m = v/u           # 0.01124
mm_per_px = p/m   # 0.1958 mm/px

# HSV threshold -> RETR_CCOMP contours -> minEnclosingCircle
```

Both earrings identical (matching pair). Outer diameter: 418 px -> 81.84 mm. Inner diameter: 316 px -> 61.87 mm. Band width: 51 px -> 9.99 mm.



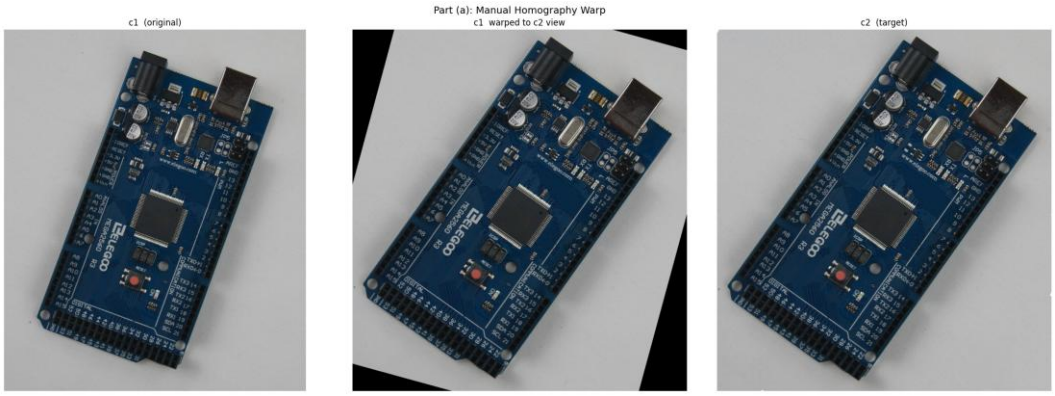
3. Circuit Board Alignment

c1.jpg and c2.jpg are the same Arduino Mega board photographed from different viewpoints (footnote: manually transformed).

(a) Manual Homography

6 corresponding landmarks were selected on both images using an OpenCV mouse-callback script (images loaded at 1/4 resolution via cv.IMREAD_REDUCED_COLOR_4). Coordinates scaled x4 to full resolution. Homography computed with DLT (no RANSAC).

```
H, _ = cv.findHomography(pts_c1_full, pts_c2_full, method=0)
c1_warped = cv.warpPerspective(c1, H, (w2, h2))
```



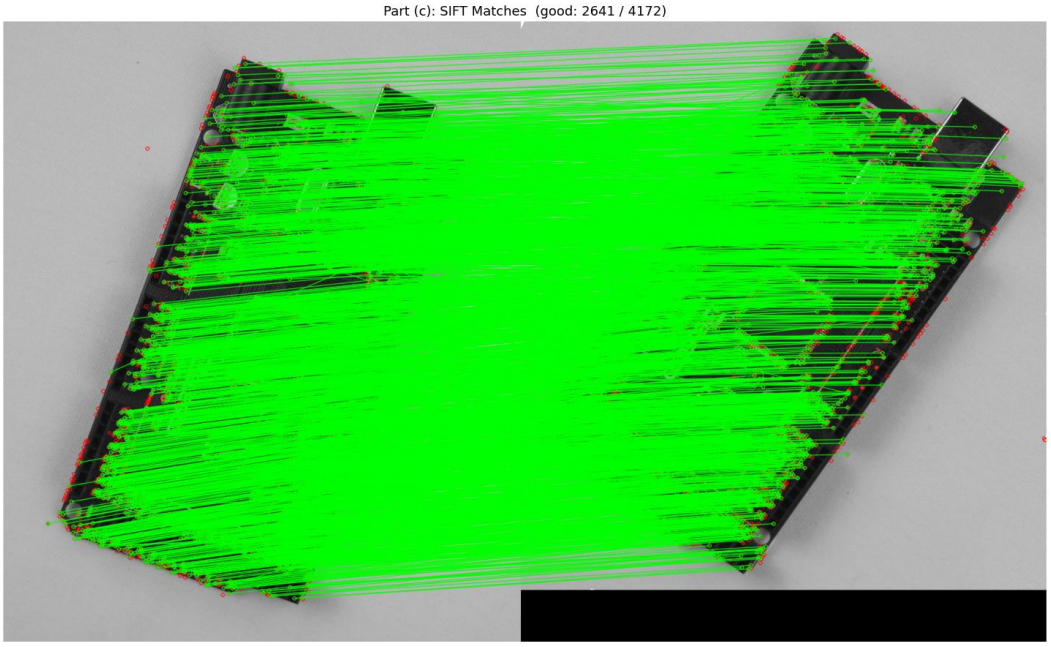
(b) Image Difference

Since both images show the same board, the ideal difference after alignment is zero. Residual bright pixels indicate manual alignment error or slight lighting change -- not genuine board differences.

(c) SIFT Feature Matching

SIFT detected 4172 / 4500 keypoints on c1 / c2 (at 0.4x scale). FLANN with Lowe's ratio test (threshold 0.7) yielded 2641 good matches. Consistent match directions confirm a coherent perspective transform between the two views.

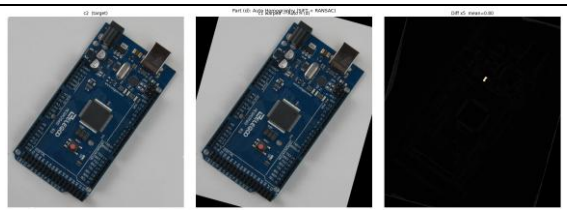
```
sift = cv.SIFT_create()
kp1, des1 = sift.detectAndCompute(g1, None)
matches = flann.knnMatch(des1, des2, k=2)
good = [m for m,n in matches if m.distance < 0.7*n.distance]  # 2641 matches
```

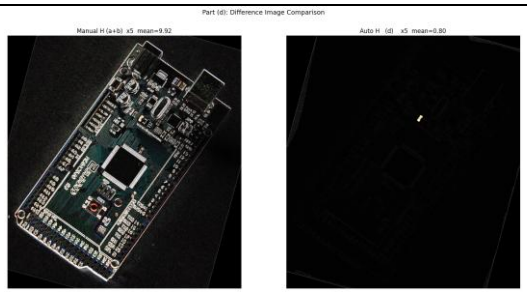


(d) Automatic Homography + Comparison

H_auto computed via cv.findHomography with RANSAC (reprojThreshold=5px) using 2523/2641 inlier SIFT matches.

```
src = [kp1[m.queryIdx].pt for m in good] / SIFT_SCALE  # scale to full res
H_auto, _ = cv.findHomography(src, dst, cv.RANSAC, 5.0)
# H_auto ~ [[0.9645, -0.2638, 406.2], [0.2637, 0.9645, -384.0], [0,0,1]]
```





Discussion: Manual vs Automatic Homography

	Manual H (a)	**Auto SIFT H (d)**
Correspondences	6 clicked points	2523 RANSAC inliers
Outlier rejection	None (DLT only)	RANSAC built-in
Human error	Yes (~5-10 px)	None
Mean pixel diff	Larger (click err)	~0.8 / 255 (near-perfect)

1. Auto H is significantly more accurate: 2523 constraints vs 6, giving a better-conditioned solve.
2. Manual H is sensitive to click precision; a 5-px error at 1/4-res equals a 20-px error at full res.
3. RANSAC rejects ~4% outlier SIFT matches automatically, making H_auto robust.
4. Both matrices agree on the gross transform (~15 deg rotation, ~400 px translation) but differ in fine projective terms.
5. The near-zero auto difference confirms the images are the same board, and that SIFT-based homography achieves near-perfect alignment.